

BAULKHAM HILLS HIGH SCHOOL

HALF YEARLY EXAMINATION

2010

YEAR 12

MATHEMATICS

EXTENSION 2

Time Allowed: 2 hours
plus 5 minutes reading time

Instructions:

- Attempt all questions.
- Do not write on this question paper. Use the answer booklets provided
- Start a new page for **each** question.
- Write your student number at the top of each page
- Use black or blue pen only
- Board approved calculators may be used
- Staple your answer pages together in ONE bundle with the QUESTIONS stapled at the back of your solutions.

Question 1 Start on a new page (10 marks)

a) Given $z = 1 - 3i$ and $w = 2 + i$

i) Express zw in the form $a + ib$ 1

ii) Find $|zw|$ and $\text{Arg}(zw)$ 3

iii) Hence, find x if $\frac{\sqrt{2}(\cos x - i \sin x)}{2 + i} = \frac{1 - 3i}{5}$ and $0 \leq x \leq \frac{\pi}{2}$. 3

b) On an Argand diagram, shade the region containing all points representing complex numbers z such that $\text{Re}(z) \leq 1$ and $0 \leq \text{Arg}(z + i) \leq \frac{\pi}{4}$. 3

Question 2 Start on a new page (10 marks)

a) i) Express $\sqrt{8 - 6i}$ in the form $a + ib$ where a and b are real. 3

ii) Hence solve $2z^2 + (1 - 3i)z - 2 = 0$ expressing the answers in the form $c + id$ where c and d are real. 2

b) The roots of $x^3 + 6x^2 + 5x - 8 = 0$ are α, β and γ .

Find

i) the value of $\alpha^3 + \beta^3 + \gamma^3$ 3

ii) the monic polynomial with roots α^2, β^2 and γ^2 . 2

Question 3 Start on a new page (10 marks)

- a) i) Show that $z = i$ is a root of $(2 - i)z^2 - (1 + i)z + 1 = 0$. 1
- ii) Find the other root in the form $z = a + ib$ where a and b are real numbers. 2
- b) i) If $P(x) = 0$ where $P(x)$ is a polynomial of degree n (where ≥ 2) has a double root at $x = \alpha$, prove that $x = \alpha$ is a single root of $P'(x)$. 2
- ii) Hence find the double root of $x^3 + x^2 - 5x + 3 = 0$. 2
- c) Given $2 + i$ and $1 - 3i$ are two roots of the equation $x^4 + bx^3 + cx^2 + dx + e = 0$ where b, c, d and e are real numbers
- i) write down the other two roots, giving a reason for your answers. 1
- ii) hence find the values of b and e only. 2

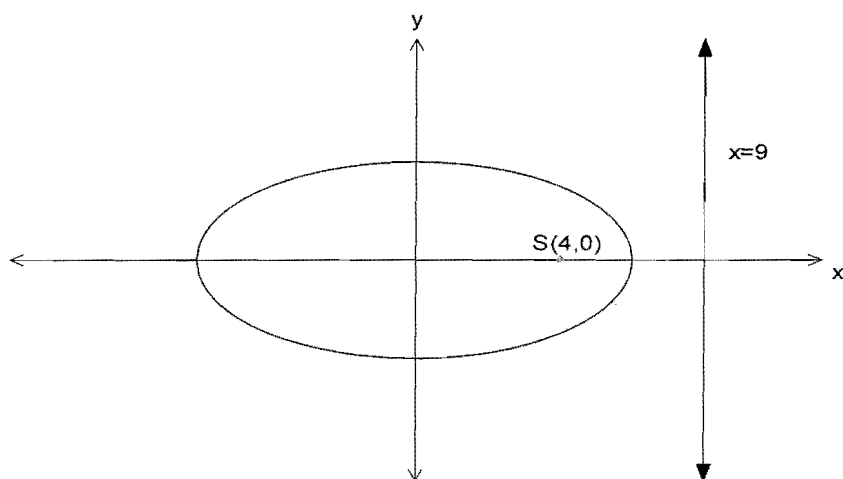
Question 4 Start on a new page (10 marks)

- a) i) If w is a complex cube root of unity, prove that $1 + w + w^2 = 0$ 2
- ii) Form an equation whose roots are $4, 1 + w, 1 + w^2$ 3
- b) An ellipse has equation $\frac{x^2}{9} + y^2 = 1$.
- i) Find the condition for the line $y = mx + c$ to be a tangent to the ellipse. 4
- ii) Hence what are the equations of the tangents with a gradient of $\frac{1}{3}$. 1

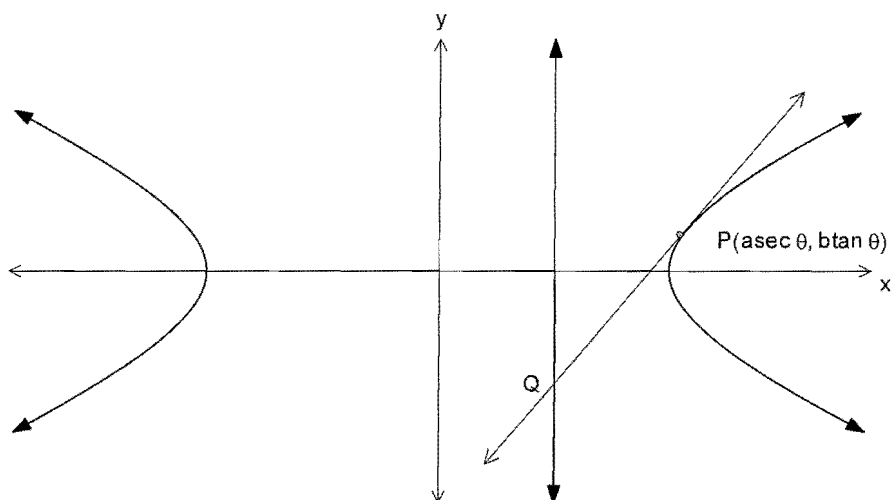
Question 5 Start on a new page (10 marks)

- a) Find the equation of the ellipse shown in the diagram below.
A focus and directrix are shown.

3



b)



- i) Derive the equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
at the point $P(a \sec \theta, b \tan \theta)$

3

- ii) This tangent meets the directrix at Q . Find the coordinates of Q .

1

- iii) If S is the focus, find the gradient of QS

1

- (iv) Prove that PQ subtends a right angle at S

2

Question 6 Start on a new page (10 marks)

a) Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$

i) Find its eccentricity 1

ii) Find the coordinates of its foci S and S' 1

iii) Find the equations of its directrices 1

iv) Sketch the ellipse showing the foci, intercepts and directrices 1

b) If β is the acute angle between the asymptotes of the hyperbola $\frac{x^2}{r^2} - \frac{y^2}{s^2} = 1$ with eccentricity e (where $r > s$)

i) find β and e in terms of r and s 3

ii) hence show that $e = \sec \frac{\beta}{2}$ 3

Question 7 Start on a new page (10 marks)

a) It is given that $|z| = z + \bar{z}$.
Sketch on an Argand diagram the locus of the point P representing z . 2

b) i) Express the roots of the equation $z^5 + 32 = 0$ in modulus/argument form. 2

ii) Hence show that

$$z^4 - 2z^3 + 4z^2 - 8z + 16 = (z^2 - 4\cos \frac{\pi}{5} z + 4)(z^2 - 4\cos \frac{3\pi}{5} z + 4) \quad 3$$

iii) Hence find the exact value of $\cos \frac{\pi}{5}$ in simplest surd form. 3

Question 8 **Start a new page**

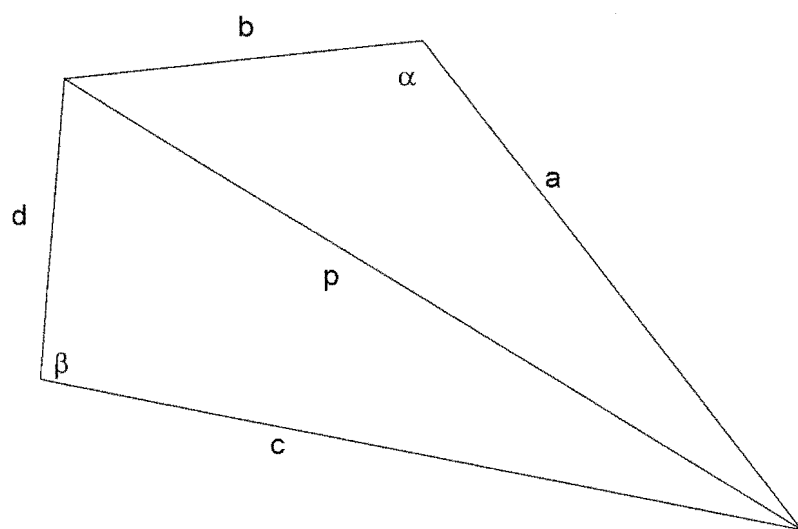
- a) Two of the roots of the quartic $6x^4 - 13x^3 - 90x^2 + 208x - 96 = 0$ are equal in magnitude but opposite in sign.

The other two roots are reciprocals of each other.

Find the four roots.

3

- b) Four lengths, a , b , c and d are joined together to form a quadrilateral.



- i) Find an expression for the area, A , of the quadrilateral.

1

- ii) Prove that $a^2 + b^2 - c^2 - d^2 = 2ab \cos \alpha - 2cd \cos \beta$

1

- iii) Show that $\frac{dA}{d\alpha} = \frac{1}{2} \left(ab \cos \alpha + cd \cos \beta \frac{d\beta}{d\alpha} \right)$ and that

2

$$\frac{d\beta}{d\alpha} = \frac{ab \sin \alpha}{cd \sin \beta}$$

- iv) Prove that the greatest area occurs when the quadrilateral is cyclic

3

End of Paper

$$1a) i) (1-3i)(2+i) = 2+i-6i-3i^2 \\ = 5-5i \quad \checkmark$$

$$ii) |5-5i| = \sqrt{25+25} \\ = 5\sqrt{2} \quad \checkmark$$

$$\arg(5-5i) = \tan^{-1}(-1) \quad \checkmark \quad \frac{3}{4} \text{ for } \frac{\pi}{4}$$

$$= -\frac{\pi}{4} \quad \checkmark$$

$$iii) \frac{\sqrt{2}(\cos \lambda - i \sin \lambda)}{2+i} = \frac{1-3i}{5}$$

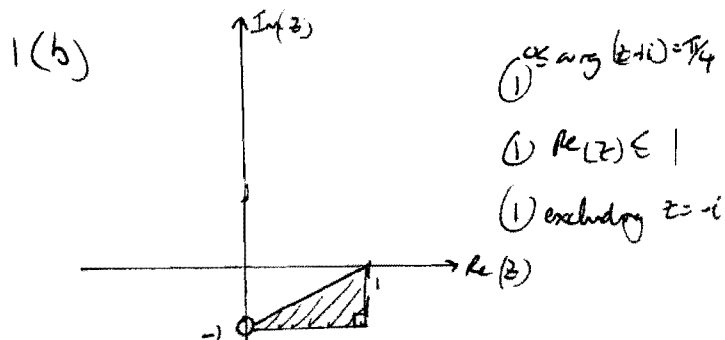
$$\sqrt{2}(\cos \lambda - i \sin \lambda) = \frac{5-5i}{5} \quad \checkmark$$

$$\sqrt{2}(\cos \lambda - i \sin \lambda) = 1-i \quad \checkmark$$

$\therefore \sqrt{2} \cos \lambda = 1 \quad -\sqrt{2} \sin \lambda = -1$ equating real and imaginary parts

$$\therefore \tan \lambda = 1$$

$$\lambda = \frac{\pi}{4} \quad \checkmark$$



$$2a) i) \sqrt{8-6i} = x+iy$$

$$8-6i = x^2-y^2+2xyi$$

$$\therefore \left. \begin{aligned} 8 &= x^2-y^2 \\ -6 &= 2xy \end{aligned} \right\} \quad \checkmark$$

$$y = \frac{-3}{x}$$

$$\therefore 8 = x^2 - \frac{9}{x^2}$$

$$8x^2 = x^4 - 9$$

$$x^4 - 8x^2 - 9 = 0$$

$$(x^2-9)(x^2+1) = 0 \quad \checkmark$$

$$x^2 = 9 \quad x^2+1=0$$

$$x = \pm 3 \quad \text{no real sol for } x \therefore y = \mp i$$

$$\therefore \pm(3-i) = \sqrt{8-6i}$$

$$(ii) 2z^2 + (1-3i)z - 2 = 0$$

$$z = \frac{-(1-3i) \pm \sqrt{(1-3i)^2 + 4 \times 2 \times 2}}{4} \quad \checkmark$$

$$z = \frac{-(1-3i) \pm \sqrt{1-6i-9+16}}{4}$$

$$z = \frac{-(1-3i) \pm \sqrt{8-6i}}{4}$$

$$z = \frac{-1+3i \pm (3-i)}{4}$$

$$z = \frac{2+2i}{4}, \frac{-4+4i}{4}$$

$$z = \frac{1}{2} + \frac{i}{2}, -1+i \quad \checkmark$$

$$2b) i) \alpha^3 = -6\alpha^2 - 5\alpha + 8$$

$$\beta^3 = -6\beta^2 - 5\beta + 8$$

$$\gamma^3 = -6\gamma^2 - 5\gamma + 8$$

$$\alpha^3 + \beta^3 + \gamma^3 = -6(\alpha^2 + \beta^2 + \gamma^2) - 5(\alpha + \beta + \gamma) + 24 \quad \checkmark$$

$$\text{Now } \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= (-6)^2 - 2 \times 5$$

$$= 26$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = -6 \times 26 - 5 \times -6 + 24$$

$$= -162 \quad \checkmark$$

$$2b) ii) \text{ let } y = x^2 \therefore \text{ roots are } \alpha^2, \beta^2, \gamma^2$$

$$\text{ie } x = \sqrt{y}$$

$$y\sqrt{y} + 6y + 5\sqrt{y} - 8 = 0$$

$$[\sqrt{y}(y+5)]^2 = [-6y+8]^2$$

$$y(y^2+10y+25) = 36y^2-96y+64$$

$$y^3-26y^2+121y-64=0$$

$$\therefore \text{Eqn is } x^3-26x^2+121x-64=0 \quad \checkmark$$

$$3 a(i) P(z) = (2-i)z^2 - (1+i)z + 1$$

$$P(i) = (2-i)(-1) - i(1+i) + 1$$

$$= -2 + i - i + 1 + 1$$

$$= 0$$

$\therefore z = i$ is a root ✓

(ii) $\therefore z - i$ is a factor. By inspection,
 $(2-i)z^2 - (1+i)z + 1 = (z-i)((2-i)z + 1)$ ✓

$\therefore$ other root is $(2-i)z = 1$

$$z = \frac{-1}{2-i} \cdot \frac{2+i}{2+i}$$

$$z = \frac{1-2i}{5}$$

$$= \frac{1}{5} - \frac{2i}{5}$$
 ✓

$$3(b) \text{ let } P(x) = (x-\alpha)^2 Q(x)$$
 ✓

$$P'(x) = Q(x) \cdot 2(x-\alpha) + (x-\alpha)^2 Q'(x)$$

$$P'(x) = (x-\alpha)(2Q(x) + (x-\alpha)Q'(x))$$

$$P'(\alpha) = 0 \cdot (T(x))$$

$$= 0$$

$\therefore x = \alpha$ is also a root of $P'(x)$

$$(ii) \text{ let } P(x) = x^3 + x^2 - 5x + 3$$

$$P'(x) = 3x^2 + 2x - 5$$

$$P'(x) = (3x+5)(x-1)$$

$$\therefore P'(x) = 0 \rightarrow x = 1, -\frac{5}{3}$$
 ✓

$$\text{Test in } P(x) \quad P(1) = 1 + 1 - 5 + 3 = 0$$

$\therefore x = 1$ is the double root ✓

3 c(i) $2-i$ and $1+3i$ since coefficients are real, roots are in conjugate pairs

(ii) sum of roots

$$(2-i) + (2+i) + (1+3i) + (1-3i) = -b$$

$$b = -6$$

$$b = -6$$
 ✓

$$\text{Product } (2-i)(2+i)(1-3i)(1+3i) = c$$

$$5 \times 10 = c$$

$$c = 50$$
 ✓

$\therefore b = -6$ and $c = 50$

$$4 a(i) \text{ for } z^3 - 1 = 0$$

$$z^3 - 1 = (z-1)(z^2 + z + 1)$$

If w is a complex root

$$w^3 - 1 = (w-1)(w^2 + w + 1)$$
 ✓

But $w \neq 1$ (as it is complex)

$$\therefore 0 = w^2 + w + 1$$
 ✓

$$(ii) \text{ sum of roots} = 4 + 1 + w + (-w)$$

$$\text{at a time} = 5$$

sum two at a time:

$$4(1+w) + 4(1+w^2) + (1+w)(1+w^2)$$
 ✓ for 1 correct expression

$$= 4 + 4w + 4 + 4w^2 + (1+w)(-w)$$

$$= 4 - w - w^2$$
 ✓ for 2 correct expressions

$$= 5$$

$$\text{three at a time } 4(1+w)(-w)$$

$$= 4(-w - w^2)$$

$$= 4$$

$$\therefore \text{Eqn is } x^3 - 5x^2 + 5x - 4 = 0 \text{ (or equivalent)} ✓$$

$$4(b)(i) \frac{x^2}{a} + (mx+c)^2 = 1$$
 ✓

$$\frac{x^2}{a} + m^2 x^2 + 2mcx + c^2 = 1$$

$$x^2 + 9m^2 x^2 + 18mcx + 9c^2 - 9 = 0$$

$$(9m^2 + 1)x^2 + 18mcx + 9c^2 - 9 = 0$$
 ✓

To be a tangent $\Delta = 0$

$$(18mc)^2 - 4(9m^2 + 1)9(c^2 - 1) = 0$$
 ✓

$$36(9m^2 c^2 - (9m^2 c^2 - 9m^2 + c^2 - 1)) = 0$$

$$9m^2 = c^2 - 1$$
 ✓

(ii) when $m = \frac{1}{3}$

$$1 = c^2 - 1$$

$$c = \pm \sqrt{2}$$

$$\therefore y = \frac{x}{3} \pm \sqrt{2}$$
 ✓

5a) Focus $(ae, 0)$ Directrix $x = \frac{a}{e}$
 $ae = 4$ $\frac{a}{e} = 9 \rightarrow e = \frac{a}{9}$

$$\frac{a^2}{9} = 4$$

$$a^2 = 36 \rightarrow a = 6$$

$$\therefore e = \frac{2}{3}$$

$$b^2 = a^2(1 - e^2)$$

$$b^2 = 36\left(1 - \frac{4}{9}\right)$$

$$b^2 = 20$$

$$\therefore \text{Eqn is } \frac{x^2}{36} + \frac{y^2}{20} = 1$$

5b) i) Different. w.r.t. x

$$\frac{2x}{a^2} - \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x b^2}{a^2 y}$$

At P, $m = \frac{a \sec \theta b^2}{a^2 b \tan \theta}$

Eqn of tangent

$$y - b \tan \theta = \frac{a \sec \theta b^2}{a^2 b \tan \theta} (x - a \sec \theta)$$

$$\frac{y \tan \theta}{b} - \frac{b \tan \theta}{b} = \frac{x \sec \theta}{a} - \sec^2 \theta$$

$$\therefore \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1 \text{ is eqn.}$$

5 (ii) At directrix $x = \frac{a}{e}$

$$\frac{\frac{a}{e} \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

$$\frac{\sec \theta}{e} - 1 = \frac{y \tan \theta}{b}$$

$$y = \frac{b}{\tan \theta} \left(\frac{\sec \theta - e}{e} \right)$$

$$\therefore Q \text{ is } \left(\frac{a}{e}, \frac{b(\sec \theta - e)}{e \tan \theta} \right) \text{ or equivalent}$$

$$\begin{aligned} \text{d(iii)} m_{QS} &= \frac{b(\sec \theta - e)}{e \tan \theta} - 0 \\ &= \frac{b(\sec \theta - e)}{\tan \theta (a - ae^2)} \end{aligned}$$

$$\begin{aligned} \text{iv) } m_{PS} &= \frac{b \tan \theta - 0}{a \sec \theta - ae} \\ &= \frac{b \tan \theta}{a(\sec \theta - e)} \end{aligned}$$

$$\begin{aligned} m_{PS} \cdot m_{QS} &= \frac{b \tan \theta}{a(\sec \theta - e)} \cdot \frac{b(\sec \theta - e)}{\tan \theta a(1 - e^2)} \\ &= \frac{b^2}{a^2(1 - e^2)} \\ &= \frac{b^2}{-a^2(e^2 - 1)} \end{aligned}$$

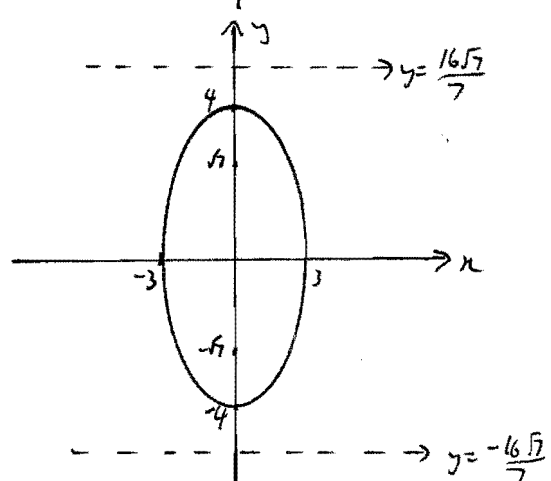
$$= -1 \text{ as } b^2 = a^2(e^2 - 1)$$

$\therefore PS \perp QS$

6(a)(i) $9 = 16(1 - e^2)$
 $\frac{9}{16} = 1 - e^2$
 $e^2 = \frac{7}{16}$
 $e = \frac{\sqrt{7}}{4}$

(ii) Foci are: $(0, \pm be)$
 $= (0, \pm 4 \times \frac{\sqrt{7}}{4})$
 $= (0, \pm \sqrt{7})$

(iii) $y = \pm \frac{b}{e}$
 $= \pm \frac{4}{\frac{\sqrt{7}}{4}}$
 $= \pm \frac{16}{\sqrt{7}}$
 $y = \pm \frac{16\sqrt{7}}{7}$



$$(b) \quad s^2 = r^2(e^2 - 1)$$

$$\frac{s^2}{r^2} = e^2 - 1$$

$$e^2 = \frac{s^2 + r^2}{r^2}$$

$$e = \frac{\sqrt{s^2 + r^2}}{r} \quad (\text{since } e > 0) \quad \checkmark$$

Asymptotes are $y = \pm \frac{s}{r}x$

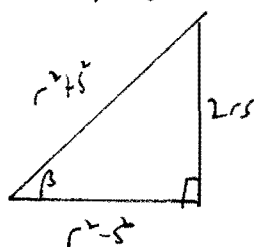
$$\tan \beta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{s}{r} - \left(-\frac{s}{r}\right)}{1 + \left(-\frac{s^2}{r^2}\right)} \right|$$

$$= \left| \frac{\frac{2s}{r}}{\frac{r^2 - s^2}{r^2}} \right|$$

$$\tan \beta = \left| \frac{2sr}{r^2 - s^2} \right|$$

$$\tan \beta = \frac{2rs}{r^2 - s^2} \quad \text{as } r > s \therefore \beta = \tan^{-1} \left(\frac{2rs}{r^2 - s^2} \right)$$



$$\cos \beta = \frac{r^2 - s^2}{r^2 + s^2} \quad \checkmark$$

$$\cos(2\beta) = 2 \cos^2 \beta - 1 \quad \checkmark$$

$$\frac{r^2 - s^2}{r^2 + s^2} = 2 \cos^2 \beta - 1$$

$$\frac{r^2 - s^2}{r^2 + s^2} = 2 \cos^2 \beta$$

$$\frac{2rs}{r^2 + s^2} = \cos^2 \beta$$

$$\sec \beta = \frac{\sqrt{r^2 + s^2}}{r}$$

$$\sec \beta = e$$

$$7 (a) \quad |z| = z + \bar{z} \quad \text{let } z = x + iy$$

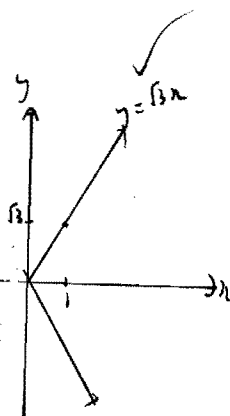
$$\sqrt{x^2 + y^2} = 2x \quad (x \geq 0)$$

$$x^2 + y^2 = 4x^2$$

$$3x^2 - y^2 = 0$$

$$y^2 = 3x^2$$

$$y = \pm \sqrt{3}x \quad (x \geq 0)$$



$$7b(i) \quad z^5 = +32(\cos \pi + 2k\pi + i \sin \pi + 2k\pi) \quad \checkmark$$

$$z = 2 \cos\left(\frac{\pi + 2k\pi}{5}\right) + i \sin\left(\frac{2k\pi + \pi}{5}\right) \quad k=0,1,2,3,4$$

$$z_1 = 2\left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}\right)$$

$$z_2 = 2\left(\cos\left(\frac{3\pi}{5}\right) + i \sin\left(\frac{3\pi}{5}\right)\right)$$

$$z_3 = 2(\cos \pi + i \sin \pi) = -2$$

$$z_4 = 2\left(\cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5}\right) = 2 \operatorname{cis}\left(-\frac{3\pi}{5}\right)$$

$$z_5 = 2\left(\cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}\right) = 2 \operatorname{cis}\left(-\frac{\pi}{5}\right)$$

$$(ii) \quad z^5 + 32 = (z+2)(z^4 - 2z^3 + 4z^2 - 8z + 16)$$

Roots of $z^5 + 32 = 0$ are z_1, z_2, z_3, z_4, z_5 .

$\therefore$ Roots of $z^4 - 2z^3 + 4z^2 - 8z + 16 = 0$ are complex $\checkmark$

roots of $z^5 + 32 = 0$ i.e. $z_1, z_2, z_4 = \bar{z}_2, z_5 = \bar{z}_1$

$$z^4 - 2z^3 + 4z^2 - 8z + 16 = (z - z_1)(z - \bar{z}_1)(z - z_2)(z - \bar{z}_2) \quad \checkmark$$

$$z_1 + \bar{z}_1 = 4 \cos \frac{\pi}{5} \quad z_1 \bar{z}_1 = 4 \quad z_2 + \bar{z}_2 = 4 \cos \frac{3\pi}{5} \quad z_2 \bar{z}_2 = 4$$

$$z^4 - 2z^3 + 4z^2 - 8z + 16 = (z^2 - (z_1 + \bar{z}_1)z + z_1 \bar{z}_1)(z^2 - (z_2 + \bar{z}_2)z + z_2 \bar{z}_2)$$

$$= (z^2 - 4 \cos \frac{\pi}{5} z + 4)(z^2 - 4 \cos \frac{3\pi}{5} z + 4)$$

$$(iii) \quad \text{equating coefficients of } z:$$

$$-8 = -16 \cos \frac{\pi}{5} - 16 \cos \frac{3\pi}{5} \quad \checkmark$$

$$1 = 2\left(\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}\right)$$

equating coeff of z^2

$$4 = 4 + 16 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} + 4$$

$$\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$$

$$\cos \frac{3\pi}{5} = \frac{-1}{4 \cos \frac{\pi}{5}}$$

$$\therefore 1 = 2 \cos \frac{\pi}{5} - \frac{1}{2 \cos \frac{\pi}{5}}$$

$$2 \cos \frac{\pi}{5} = 4 \cos^2 \frac{\pi}{5} - 1$$

$$4 \cos^2 \frac{\pi}{5} - 2 \cos \frac{\pi}{5} - 1 = 0$$

$$\cos \frac{\pi}{5} = \frac{2 \pm \sqrt{4+16}}{8}$$

$$= \frac{1 \pm \sqrt{5}}{4}$$

$$\text{since } \cos \frac{\pi}{5} > 0, \cos \frac{\pi}{5} = \frac{1 + \sqrt{5}}{4}$$

$$8(c) \text{ let roots be } \alpha, -\alpha, \beta, \frac{1}{\beta}$$

$$\text{sum of roots} = \beta + \frac{1}{\beta} = \frac{13}{6}$$

$$6\beta^2 + 6 = 13\beta$$

$$6\beta^2 - 13\beta + 6 = 0$$

$$(2\beta-3)(3\beta-2) = 0$$

$$\beta = \frac{3}{2}, \beta = \frac{2}{3}$$

product of roots

$$-\alpha^2 = -\frac{96}{6}$$

$$\alpha = \pm 4$$

$$\therefore \text{roots are } 4, -4, \frac{3}{2}, \frac{2}{3}$$

$$b(i) A = \frac{1}{2} ab \sin \alpha + \frac{1}{2} cd \sin \beta$$

(ii) using cosine rule

$$a^2 = a^2 + b^2 - 2ab \cos \alpha$$

$$\text{and } p^2 = c^2 + d^2 - 2cd \cos \beta$$

$$\therefore a^2 + b^2 - 2ab \cos \alpha = c^2 + d^2 - 2cd \cos \beta$$

$$a^2 + b^2 - c^2 - d^2 = 2ab \cos \alpha - 2cd \cos \beta$$

$$(iii) \frac{dA}{d\alpha} = \frac{1}{2} ab \cos \alpha + \frac{1}{2} cd \cos \beta \cdot \frac{d\beta}{d\alpha}$$

$$= \frac{1}{2} (ab \cos \alpha + cd \cos \beta \frac{d\beta}{d\alpha})$$

$$\frac{d(a^2 + b^2 - c^2 - d^2)}{d\alpha} = -2ab \sin \alpha + 2cd \sin \beta \cdot \frac{d\beta}{d\alpha}$$

$$0 = -2ab \sin \alpha + 2cd \sin \beta \frac{d\beta}{d\alpha}$$

$$Kab \sin \alpha = Kcd \sin \beta \frac{d\beta}{d\alpha}$$

$$\frac{d\beta}{d\alpha} = \frac{ab \sin \alpha}{cd \sin \beta}$$

$$(iv) \frac{dA}{d\alpha} = \frac{1}{2} (ab \cos \alpha + cd \cos \beta \cdot \frac{ab \sin \alpha}{cd \sin \beta})$$

$$= \frac{1}{2} ab \left(\frac{\cos \alpha \sin \beta + \sin \alpha \cos \beta}{\sin \beta} \right)$$

$$\text{for max or min; } \frac{dA}{d\alpha} = 0$$

$$\frac{1}{2} ab \left(\frac{\sin(\alpha + \beta)}{\sin \beta} \right) = 0$$

$$\therefore \sin(\alpha + \beta) = 0$$

$$\alpha + \beta = 0 \rightarrow \text{min area}$$

$$\alpha + \beta = \pi \rightarrow \text{max area}$$

$\therefore \alpha$ and β are supplementary

$\therefore$ quadrilateral is cyclic